Circular Measure

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Converting between degrees and radians**

Convert 135° to radians, in terms of π.

$135^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{4}$.

2 **[Foundational] • Target: Arc length**

A circle has radius 8 cm. Find the length of an arc subtending an angle of $\frac{\pi}{3}$ radians at the center, using $s = r\theta$.

$s = r\theta = 8 \times \frac{\pi}{3} = \frac{8\pi}{3}$ cm.

3 **[Foundational] • Target: Sector area**

Find the area of a sector with radius 6 cm and angle $\frac{2\pi}{3}$ radians, using $A = \frac{1}{2}r^2\theta$.

$A = \frac{1}{2}(6)^2\left(\frac{2\pi}{3}\right) = \frac{1}{2}(36)\left(\frac{2\pi}{3}\right) = 12\pi$ cm^2.

4 **[Foundational] • Target: Sector area**

A sector has radius 10 cm and area 40 cm^2. Find the angle θ in radians.

$40 = \frac{1}{2}(10)^2\theta = 50\theta$, so $\theta = \frac{40}{50} = 0.8$ radians.

5 **[Foundational] • Target: Perimeter of a sector**

Find the perimeter of a sector with radius 7 cm and angle 1.2 radians.

Perimeter $= 2r + r\theta = 2(7) + 7(1.2) = 14 + 8.4 = 22.4$ cm.

6 **[Foundational] • Target: The area of a segment**

A circle has radius 5 cm. A chord subtends an angle of $\frac{\pi}{2}$ radians at the center. Find the area of the minor segment, using segment area = sector area minus triangle area.

Sector area $= \frac{1}{2}(5)^2\left(\frac{\pi}{2}\right) = \frac{25\pi}{4}$. Triangle area $= \frac{1}{2}(5)(5)\sin\left(\frac{\pi}{2}\right) = \frac{25}{2}(1) = 12.5$. Segment area $= \frac{25\pi}{4} - 12.5 \approx 19.63 - 12.5 \approx 7.13$ cm^2.

7 **[Foundational] • Target: Small-angle approximations**

Use the small-angle approximations $\sin\theta \approx \theta$ and $\cos\theta \approx 1 - \frac{\theta^2}{2}$ to approximate $\sin(0.1)$ and $\cos(0.1)$, to four decimal places.

$\sin(0.1) \approx 0.1$. $\cos(0.1) \approx 1 - \frac{0.01}{2} = 1 - 0.005 = 0.9950$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Small-angle approximations**

Use the small-angle approximation $\tan\theta \approx \theta$ to approximate $\tan(0.2)$, and compare it to the exact value $\tan(0.2) \approx 0.2027$.

$\tan(0.2) \approx 0.2$ using the small-angle approximation. The exact value is $\tan(0.2) \approx 0.2027$, so the approximation differs by only about 0.0027.

9 **[Intermediate]** • **Target: Angular speed and radians per second**

A wheel rotates at 5 radians per second. Find the angle turned through, in radians, after 8 seconds, and convert this to the number of complete revolutions.

Angle $= 5 \times 8 = 40$ radians. Number of revolutions $= \frac{40}{2\pi} \approx 6.37$ revolutions.

10 **[Intermediate]** • **Target: A well-designed word problem: a swinging pendulum's arc**

This question has two parts. A pendulum of length 1.2 meters swings through an angle of 0.3 radians on each side of vertical (a total swing angle of 0.6 radians). (a) Find the length of the arc traced by the pendulum's tip during one full swing (from one side to the other). (b) Find the area swept out by the pendulum during this swing.

(a) Arc length $= r\theta = 1.2 \times 0.6 = 0.72$ meters. (b) Area of the sector swept $= \frac{1}{2}r^2\theta = \frac{1}{2}(1.2)^2(0.6) = \frac{1}{2}(1.44)(0.6) = 0.432$ square meters.

11 **[Intermediate]** • **Target: Sector area**

For the ellipse $\frac{x^2}{81} + \frac{y^2}{9} = 1$, find the length of the major axis and the foci.

Final Answer: Major axis 18, foci $(\pm\sqrt{72}, 0)$. $a = 9 > b = 3$, so major axis $= 2a = 18$ (along x-axis). $c^2 = a^2 - b^2 = 72$, foci at $(\pm\sqrt{72}, 0)$.

12 **[Intermediate]** • **Target: Perimeter of a sector**

$x1200isinvestedatasimpleinterestrateof8\%$ per year for 5 years. Find the interest earned.

Final Answer: 480.00. Working through this modelling step: $I = \frac{Prt}{100} = \frac{(1200)(8)(5)}{100} = 480.00$.

13 **[Intermediate]** • **Target: The area of a segment**

How many distinct arrangements are there of the letters in the word LEVEL? Show your complete algebraic work.

Final Answer: 30. The word has 5 letters with L and E each repeated twice: $\frac{5!}{2!\,2!} = 30$.

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14 [Intermediate] • Target: Small-angle approximations

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)? Show your complete algebraic work.

Final Answer: 8.3 years. Solve $1.5 = (1 + 0.05)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.05)} \approx 8.3$ years.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Angular speed and radians per second • Cross-Lesson: 5.01 Trigonometry

Find the area of the polar region bounded by $r = 2$ for $0 \le \theta \le \frac{\pi}{6}$.

Final Answer: $\frac{\pi}{3}$. $A = \frac{1}{2}r^2\theta = \frac{1}{2}(2)^2(\frac{\pi}{6}) = \frac{\pi}{3}$.

16 [Challenging] • Target: A well-designed word problem: a swinging pendulum's arc • Cross-Lesson: 6.01 Differentiation & Integration

Find the reference angle for 220°.

Final Answer: 40°. 220° lies in Quadrant 3; the reference angle is 40°.

17 [Challenging] • Target: Converting between degrees and radians • Cross-Lesson: 7.01 Statistics & Probability

Solve $\cos(\theta) = -\frac{\sqrt{3}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. Find the smallest solution (in degrees).

Solutions: 150°, 210°. Smallest: 150°.

18 [Challenging] • Target: Arc length • Cross-Lesson: 8.01 Functions: Composite & Inverse

A right triangle has one leg of length 12 and hypotenuse 20. Find the length of the other leg.

By Pythagoras: $12^2 + 16^2 = 20^2$, so the unknown side is 16.

19 [Challenging] • Target: Sector area • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Solve $\sin(\theta) = \frac{\sqrt{3}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. Find the largest solution (in degrees).

Solutions: 60°, 120°. Largest: 120°.

20 [Challenging] • Target: Perimeter of a sector • Cross-Lesson: 10.01 Series & Binomial Theorem

In ΔABC, $a = 5$, $b = 7$, and $\angle C = 60^\circ$. Find side c (Law of Cosines).

Final Answer: $\sqrt{39}$. $c^2 = 5^2 + 7^2 - 2(5)(7)\cos 60^\circ \Rightarrow c = \sqrt{39}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Converting between degrees and radians

A sector of a circle of radius 12 cm has an arc length of 18 cm. Find the area of the sector.

a. $108\ \text{cm}^2$ ✓

b. $216\ \text{cm}^2$

c. $54\ \text{cm}^2$

d. $144\ \text{cm}^2$

The arc length s of a sector of radius r subtending an angle θ radians at the centre is given by $s = r\theta$. Substituting the given values gives $18 = 12\theta \Rightarrow \theta = 1.5$ radians. The area A of the sector is given by $A = \frac{1}{2}r^2\theta = \frac{1}{2}(12)^2(1.5) = 108\ \text{cm}^2$ (or directly via $A = \frac{1}{2}rs = \frac{1}{2} \times 12 \times 18 = 108\ \text{cm}^2$). Omitting the factor of $\frac{1}{2}$ yields the incorrect value of $216\ \text{cm}^2$, while.

22 [Foundational] • Target: Arc length

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?

a. **6 units** ✓

b. $6\sqrt{3}$ units

c. 12 units

d. $12\sqrt{3}$ units

We have a circle with radius $r = 6$ and a sector with a central angle of 60 degrees. The chord connecting the endpoints of the arc of this sector forms an isosceles triangle with the two radii. Since the central angle is 60 degrees and the other two angles in the triangle must be equal (because the sides opposite them are radii), each of the other angles is $(180^\circ - 60^\circ)/2 = 60^\circ$. Therefore, the triangle is equilateral. In an equilateral triangle, all sides are equal. Since two sides are radii of.

23 [Foundational] • Target: Sector area

A circle has radius 8 cm. Find the arc length subtended by an angle of 1.5 radians.

a. **12 cm** ✓

b. 6 cm

c. 24 cm

d. 9.6 cm

Arc length $= r\theta = 8 \times 1.5 = 12$ cm.

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24 [Foundational] • Target: Perimeter of a sector

A sector's perimeter is 26 cm and its radius is 8 cm. Find the angle of the sector, in radians.

a. **1.25** ✓

b. 1.625

c. 0.8

d. 2

Perimeter $= 2r + r\theta = 26 \Rightarrow 16 + 8\theta = 26 \Rightarrow \theta = 1.25$ rad.

25 [Foundational] • Target: The area of a segment

A sector of a circle has arc length 5π cm and angle 180°. Find the radius.

a. 1

b. **5** ✓

c. 2

d. 3

$s = r\theta \Rightarrow r = \frac{s}{\theta} = \frac{5\pi}{1\pi/1} = 5$ cm.

26 [Foundational] • Target: Small-angle approximations

A sector of a circle has radius 9 cm and angle 45°. Find the exact arc length.

a. $\frac{5\pi}{4}$

b. $\frac{9\pi}{4}$ ✓

c. $\frac{3\pi}{2}$

d. $\frac{7\pi}{4}$

$s = r\theta = (9)\left(\frac{1\pi}{4}\right) = \frac{9\pi}{4}$ cm.

27 [Foundational] • Target: Angular speed and radians per second

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?

a. 10.3

b. **12.1** ✓

c. 14.8

d. 8.5

We can use the Law of Cosines to solve for the unknown side BC. The Law of Cosines states that for any triangle with sides a, b, and c, and opposite angles A, B, and C respectively: $a^2 = b^2 + c^2 - 2bc\cos(A)$. In this problem, we have: side b = AC = 10, side c = AB = 14, and angle A = 52 degrees. We want to find side a = BC. Substituting the values into the formula: $a^2 = 10^2 + 14^2 - 2(10)(14)\cos(52^\circ)$ $a^2 = 100 + 196 - 280\cos(52^\circ)$ $a^2 = 296 - 280(0.61566)$ $a^2 = 296 - 172.3848$.

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28 [Foundational] • Target: A well-designed word problem: a swinging pendulum's arc

The function f is defined for all real numbers x by $f(x) = 2 - 3\sin(4x)$. What are the amplitude and the period (in radians) of f?

a. Amplitude $= 3$, Period $= \frac{\pi}{2}$ ✓

b. Amplitude $= -3$, Period $= \frac{\pi}{2}$

c. Amplitude $= 3$, Period $= 8\pi$

d. Amplitude $= 5$, Period $= \frac{\pi}{4}$

For a sinusoidal function written in the form $f(x) = c + a\sin(bx)$ (or $c - a\sin(bx)$): 1. **Amplitude:** The amplitude represents the maximum vertical distance from the horizontal centerline (midline $y = 2$) to a peak or trough. It is defined as the absolute value of the coefficient of the trigonometric ratio: Amplitude $= |-3| = 3$. Amplitude is strictly a non-negative measure of distance; it can never be negative. 2. **Period:** The fundamental period of $\sin(x)$ is 2π. For $\sin(bx)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Converting between degrees and radians

In triangle ABC, side $a = 10$ units and side $b = 15$ units. If the angle $C = 60^\circ$, find the length of side c. Round your answer to two decimal places.

a. **13.23** ✓

b. 14.70

c. 12.96

d. 15.52

We can use the Law of Cosines to find the length of side c. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab\cos C$. Given: $a = 10$, $b = 15$, and $C = 60^\circ$. Substitute the values into the formula: $c^2 = 10^2 + 15^2 - 2(10)(15)\cos 60^\circ$ $c^2 = 100 + 225 - 300 \cdot \frac{1}{2}$ $c^2 = 325 - 150$ $c^2 = 175$ $c = \sqrt{175}$ $c = \sqrt{25 \cdot 7}$ $c = 5\sqrt{7}$ To find the decimal approximation, we calculate: $c \approx 5 \cdot 2.64575$ $c \approx 13.22875$ Rounding to two decimal places.

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30 [Intermediate] • Target: Arc length

Find all solutions of $2\cos^2 x + 3\sin x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

a. $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$ ✓

b. $x = \frac{\pi}{3},\ \frac{2\pi}{3}$

c. $x = \frac{\pi}{6},\ \frac{7\pi}{6}$

d. $x = 0,\ \pi$

Solving quadratic trig equation $2\cos^2 x + 3\sin x - 3 = 0$ yields solutions $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$.

31 [Intermediate] • Target: Sector area

Given $\mathbf{a} = (2,\ -6,\ -9)$ and $\mathbf{b} = (2, 3, 6)$, find $\cos\theta$, where θ is the angle between them.

a. $\frac{9}{77}$

b. $\frac{68}{77}$

c. $-\frac{34}{9}$

d. $-\frac{68}{77}$ ✓

$\mathbf{a} \cdot \mathbf{b} = -68$, $|\mathbf{a}| = 11$, $|\mathbf{b}| = 7$. $\cos\theta = \frac{-68}{11 \times 7} = -\frac{68}{77}$.

32 [Intermediate] • Target: Perimeter of a sector

Find the angle between $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle 0, 1 \rangle$.

a. 135°

b. 60°

c. 90° ✓

d. 120°

$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = 0 \Rightarrow \theta = 90^\circ$.

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33 [Intermediate] • Target: The area of a segment

A particle moves in the xy-plane. Its velocity vector $\mathbf{v}(t)$ at time t is given by $\mathbf{v}(t) = \langle 3t^2 - 6t, e^t - 1 \rangle$, where $t \geq 0$. The initial position of the particle at $t = 0$ is $\mathbf{r}(0) = \langle 5, -2 \rangle$. Find the position vector $\mathbf{r}(t)$ of the particle at time t. Then, determine the value of t for which the particle's speed is minimized and state the minimum speed.

a. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{2}$ at $t = 1$

b. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{5}$ at $t = 0$

c. $\mathbf{r}(t) = \langle 3t^2 - 6t + 5, e^t - t - 2 \rangle$, minimum speed 0 at $t = 0$

d. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$**, minimum speed** $\sqrt{3}$ **at** $t = 0$ ✓

To find the position vector $\mathbf{r}(t)$, we integrate the velocity vector $\mathbf{v}(t)$ with respect to t. $\mathbf{r}(t) = \int \mathbf{v}(t)dt = \int \langle 3t^2 - 6t, e^t - 1 \rangle dt$ $\mathbf{r}(t) = \left\langle \int (3t^2 - 6t)dt, \int (e^t - 1)dt \right\rangle$ $\mathbf{r}(t) = \langle t^3 - 3t^2 + C_1, e^t - t + C_2 \rangle$ Using the initial condition $\mathbf{r}(0) = \langle 5, -2 \rangle$: For the x-component: $0^3 - 3(0)^2 + C_1 = 5 \Rightarrow C_1 = 5$ For the y-com.

34 [Intermediate] • Target: Small-angle approximations

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.

a. 44

b. 45 ✓

c. 50

d. 54

Let the arithmetic progression (u_n) have first term a and common difference d. Then: $u_1 = a, \quad u_4 = a + 3d, \quad u_6 = a + 5d.$ Let the geometric progression (v_n) have first term v_1 and common ratio r. Since $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$, we have: $v_1 = a, \quad v_2 = ar, \quad v_3 = ar^2.$ Equating terms: 1) $a + 3d = ar \Rightarrow 3d = a(r - 1) \Rightarrow d = \frac{a(r-1)}{3}$. 2) $a + 5d = ar^2$. Substitute d into the second equation.

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35 [Intermediate] • Target: Angular speed and radians per second

Let the complex number z_1 be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, z_2, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of θ_2 in radians, given that $0 \le \theta_2 < 2\pi$.

a. $\frac{\pi}{2}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{6}$

We are given two complex numbers in polar form: $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. We are provided with $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. For z_2, we are given $r_2 = 2$. The product of two complex numbers in polar form is given by $z_1 \cdot z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$. We are given that $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$, where $\alpha = \frac{5\pi}{6}$. Equating the magnitudes, we.

36 [Intermediate] • Target: A well-designed word problem: a swinging pendulum's arc

Consider a geometric progression where the sum of the first n terms is given by $S_n = \frac{3(2^n - 1)}{2}$. A second sequence, denoted by a_n, is defined such that $a_n = S_n - S_{n-1}$ for $n \ge 2$, and $a_1 = S_1$. Find the value of $a_5 - a_3$.

a. 24

b. 18

c. 48

d. **36** ✓

The sum of the first n terms of the geometric progression is $S_n = \frac{3(2^n - 1)}{2}$. The n-th term of the sequence, a_n, can be found using the relation $a_n = S_n - S_{n-1}$ for $n \ge 2$, and $a_1 = S_1$. First, let's find a_1: $a_1 = S_1 = \frac{3(2^1 - 1)}{2} = \frac{3(1)}{2} = \frac{3}{2}$. Now, let's find a general formula for a_n for $n \ge 2$: $S_{n-1} = \frac{3(2^{n-1} - 1)}{2}$. $a_n = S_n - S_{n-1} = \frac{3(2^n - 1)}{2} - \frac{3(2^{n-1} - 1)}{2}$.

37 [Intermediate] • Target: Converting between degrees and radians

The continuous function f is defined for all $x \in \mathbb{R}$ by $f(x) = 2^x + 3x - 11$. Between which pair of consecutive integers does a root of the equation $f(x) = 0$ lie?

a. 2 **and** 3 ✓

b. 1 and 2

c. 3 and 4

d. 0 and 1

To determine the interval containing a root of a continuous function $f(x)$, we use the change-of-sign rule (Intermediate Value Theorem). If $f(a)$ and $f(b)$ have opposite signs, then at least one real root of $f(x) = 0$ lies in the interval (a, b). Evaluate $f(x)$ at consecutive integers: - For $x = 0$: $f(0) = 2^0 + 3(0) - 11 = 1 + 0 - 11 = -100$ - For $x = 4$: $f(4) = 2^4 + 3(4) - 11 = 16 + 12 - 11 = 17 > 0$ Since $f(2) = -10$, there is a change of sign between $x = 2$ and $x = 3$. Because f is continuous.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Arc length • Cross-Lesson: 8.01 Functions: Composite & Inverse

Relative to a fixed origin O, the points A and B have position vectors $\vec{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\vec{AB}$.

a. 7 ✓

b. $\sqrt{31}$

c. $\sqrt{69}$

d. 49

To find the vector $\vec{AB}$, subtract the position vector of A from that of B: $\vec{AB} = \vec{OB} - \vec{OA} = (3-1)\mathbf{i} + (2-5)\mathbf{j} + (4-(-2))\mathbf{k} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$. The magnitude (length) of $\vec{AB}$ is: $|\vec{AB}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4+9+36} = \sqrt{49} = 7$. Common misconceptions: - Incorrectly taking $(-3)^2 = -9$ leads to $\sqrt{4-9+36} = \sqrt{31}$. - Adding th.

39 [Challenging] • Target: Sector area • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The function f is defined for all real numbers by $f(x) = x^3 + 3x - 5$. Between which pair of consecutive integers does the unique real root of the equation $f(x) = 0$ lie?

a. 1 **and** 2 ✓

b. 0 and 1

c. 2 and 3

d. -1 and 0

To locate the root using the change of sign method, evaluate $f(x)$ at consecutive integer values: $f(0) = (0)^3 + 3(0) - 5 = -50$ Since $f(x)$ is a continuous polynomial function and there is a sign change between $x = 1$ (where $f(1)0$), by the Intermediate Value Theorem, a root of $f(x) = 0$ lies between 1 and 2 . Checking the other intervals: - On $[0, 1]$, $f(0) = -5$ and $f(1) = -1$ (no sign change). - On $[2, 3]$, $f(2) = 9$ and $f(3) = 31$ (no sign change). - On $[-1, 0]$, $f(-1) = -9$ and $f(0) = -5$.

40 [Challenging] • Target: Perimeter of a sector • Cross-Lesson: 10.01 Series & Binomial Theorem

A car purchased for $x20000depreciatesatarateof20\%$ per year (reducing balance). Find its value after 3 years, correct to 2 decimal places.

a. **10240.00** ✓

b. 9640.00

c. 10540.00

d. 9940.00

Working through this modelling step: $A = P(1 - \frac{r}{100})^n = (20000)(1 - \frac{20}{100})^3 \approx 10240.00$.

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41 [Challenging] • Target: The area of a segment • Cross-Lesson: 11.01 Kinematics

$x1500isinvestedatacompoundinterestrateof7\%$ per year, compounded annually, for 5 years. Find the value of the investment, correct to 2 decimal places.

a. **2103.83** ✓

b. 2063.83

c. 2143.83

d. 2023.83

Working through this modelling step: $A = P\left(1 + \frac{r}{100}\right)^n = (1500)\left(1 + \frac{7}{100}\right)^5 \approx 2103.83$.

42 [Challenging] • Target: Small-angle approximations • Cross-Lesson: 12.01 Forces & Equilibrium

Find a positive coterminal angle with 400° in $[0^\circ, 360^\circ)$.

a. 30

b. **40** ✓

c. 31

d. 400

Applying Trigonometry principles, $400^\circ - 360^\circ = 40^\circ$. Therefore, the verified correct option is 40.

43 [Challenging] • Target: Angular speed and radians per second • Cross-Lesson: 13.01 Newton's Laws

Given $z = r(\cos 90^\circ + i\sin 90^\circ)$, find the principal argument (in degrees, $0 \le \theta < 360$) of z^3.

a. **270** ✓

b. 0

c. 274

d. 90

By De Moivre's theorem, $\arg(z^3) = n\theta = 3 \times 90^\circ = 270^\circ \equiv 270^\circ \pmod{360}$.

44 [Challenging] • Target: A well-designed word problem: a swinging pendulum's arc • Cross-Lesson: 14.01 Energy, Work & Power

Find the reference angle of 120°.

a. 120

b. **60** ✓

c. 31

d. 30

Applying Trigonometry principles, Reference angle $= 60^\circ$. Therefore, the verified correct option is 60.

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45 [Challenging] • Target: Converting between degrees and radians • Cross-Lesson: 15.01 Representation of Data

Find the area enclosed by the curve $y = x^2$, the x-axis, and the line $x = 3$, for $0 \le x \le 3$.

a. **9** ✓

b. 5

c. 4

d. 6

Working through this calculus step: Area $= \int_0^3 x^2\,dx = [\frac{1}{3}x^3]_0^3 = 9$.

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